

Zhuo (Prince) Niu

Product Designer & AI Product Developer | Human Factors Engineering

Zhuo.Niu@tufts.edu · (765) 838-9636 · Medford, MA
linkedin.com/in/zhuoniu110 · github.com/princeniu · www.princeniu.com

SUMMARY

- Human Factors Engineering M.S. student at Tufts with a Computer Engineering background, focused on human-AI interaction.
- Independently designed, built, and released **Helmline on the App Store** and SayIt for macOS, from interaction design to native implementation and distribution.
- Combine stakeholder research, task analysis, and usability testing with hands-on prototyping and software development.

EDUCATION

Tufts University — Master of Science, Human Factors Engineering Sep 2025 - Expected Dec 2026
Medford, MA

Relevant Courses: Human-Machine System Design, Computer Interface Design, Advanced Research Methods, Applied Behavioral Statistics, Human Factors Product Design, Use-Related Risk Management

Purdue University — Bachelor of Science, Computer Engineering Aug 2020 – Dec 2024
West Lafayette, IN

- Relevant Courses:** Advanced C Programming, Data Structures, Microprocessor Systems and Interfacing, Computer Graphics, Operating Systems, Data Science, Artificial Intelligence, Computer Security, Data Mining

WORK EXPERIENCE

UI/UX Designer — Porsche (Porsche Asia Pacific) Jun 2024 – Sep 2024
Hong Kong, China SAR

- Collaborated with **cross-functional design and engineering teams** to prototype UI components for Porsche’s in-car system, supporting both touch and voice-based navigation.
- Developed **high-fidelity Figma prototypes and component specs** to clarify interaction details and streamline frontend handoff.
- Presented **design rationale and interaction flows** to align team members on usability goals, visual consistency, and technical feasibility.
- Optimized UI responsiveness and micro-interactions through iterative feedback, contributing to a **30% reduction in load time**.

User System Developer — China GPI Tech Jun 2023 – Aug 2023
Shanghai, China

- Collaborated with **product and engineering teams** to design multimodal UX flows integrating voice, touch, and GUI interactions for an AI-driven user system.
- Mapped **user interaction scenarios** and refined NLP voice commands, improving speech recognition accuracy to **95%**.
- Presented and documented **input transition logic** to align developers on smoother handoff between voice and touch commands.

TECHNICAL SKILLS

UI & Interaction Design: Figma, User Research, Task Analysis, Cognitive Task Analysis, Usability Testing, User Flows, Prototyping, Design Systems, Human Interface Guidelines, Material Design

Programming & App Development: Python, Swift, SwiftUI, JavaScript, TypeScript, React, React Native, HTML, CSS, Tailwind CSS, C/C++

AI & Agentic Tools: Prompt Engineering, AI-Assisted Coding, OpenAI Codex, Claude Code, RAG Systems, Dify, AI Agent Workflows, Human-AI Interaction, Whisper / Speech-to-Text

Tools, APIs & Data: Git, GitHub, Firebase, Expo, Xcode, RESTful APIs, GraphQL, SQL, PostgreSQL, FastAPI

PROJECTS

Helmline | Native iOS Client for AI Agents

2026

Portfolio Case Study

- Independently designed, built, and **released a SwiftUI iPhone app on the App Store** for monitoring and controlling self-hosted Hermes Agent sessions on Mac and VPS hosts.
- Designed **summary-to-detail execution views** and explicit approval scopes; required Face ID or device passcode for positive approvals while keeping agent execution on the host.
- Integrated **Share Extension, voice capture, Widget, and Live Activity**; kept Gateway credentials and task context isolated across connections. Delivered English and Simplified Chinese support.
- **Tools:** Swift, SwiftUI, Xcode, Keychain, App Groups, WidgetKit, ActivityKit

TrekAssist

Sep 2025 - Dec 2025

Portfolio Case Study

- Collaborated on a **Figma prototype** for a solar-powered e-ink national park guide designed for Yellowstone visitors under low-power, **1 Hz refresh-rate display constraints**.
- Conducted **usability testing with 9 participants**, tracking task success, completion time, errors, attempts, and ease ratings across six outdoor-use scenarios.
- Translated test findings into **design updates**, improving alert differentiation, battery-setting access, map readability, and persistent emergency access.
- Designed **safety-centered interaction flows** for weather warnings, ranger SOS confirmation, trail navigation, guided tours, visitor safety tips, and low-power settings.
- **Tools Used:** Figma, FigJam, Usability Testing, User Flows, Prototyping, Excel

SayIt

Mar 2026 - Apr 2026

Portfolio Case Study

- Developed a **production-ready macOS menu bar app** for voice-to-text transcription, featuring global hotkey recording, live status feedback, and automatic clipboard copy.
- Integrated **Apple Speech and on-device Whisper via whisper.cpp**, enabling privacy-focused transcription with configurable engines and local model support.
- Designed and implemented a polished **SwiftUI/AppKit interface** with microphone selection, audio level visualization, HUD notifications, and bilingual localization.
- Shipped a **downloadable v1.0.5 release** through GitHub Releases with DMG packaging, Developer ID signing, notarization, Sparkle appcast updates, and versioned release workflow.
- **Tools Used:** Swift, SwiftUI, AppKit, AVFoundation, Speech Framework, CoreAudio, whisper.cpp, Xcode, GitHub

KnowledgeOS

May 2026

Portfolio Case Study

- Built a **personal knowledge system** where an AI agent handles classification, deduplication, and review, replacing manual note-taking with automated organization. Scaled to **100+ Markdown files, 400+ wikilinks**, and 0 broken links.
- Designed the rules that decide what gets saved, how it's categorized, and when duplicates merge, then automated the pipeline with **3 cron jobs** running daily capture, weekly synthesis, and maintenance audits.
- Released the system as an **open-source Starter Vault template** on GitHub (public template repo), with setup guides, customization docs, example workflows, and validation scripts.
- **Tools Used:** Python, Markdown, Obsidian, Git, GitHub, Hermes Agent, Cron Automation, Agent Workflow Design

PROFESSIONAL CERTIFICATES

Google UX Design Professional Certificate — Coursera · 2024

Skills Acquired: User Research, Wireframing, Prototyping, Usability Testing

Meta iOS Developer Professional Certificate — Coursera · 2024

Skills Acquired: Swift, iOS Development, React Native, Core Data